

Banking, tracking, and sharing of materials

Risk assessment

▷ Wear gloves, safety glasses, lab coat



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This is a bench card. Full protocol available online.

>> **Setting up the inventory system**

- (1.) **Critical:** Let the inventory issue identifiers as consecutive numbers with a material-type prefix. When the system hands out the next free number, no one accidentally picks a used one. 
- (2.) Prioritize putting the identifier on the label, nothing else matters more. With no room for much detail, the inventory becomes the natural place to look and to update.
- (3.) Keep the inventory simple enough that adding an entry is faster than skipping it. Back it up and keep distributed copies; losing the inventory is far more expensive than preventing that loss. 
- (4.) **Optional:** For large inventories, barcode vials or boxes to speed up handling.
- (5.) Index every storage box and maintain digital box maps. Number boxes consecutively. 

Quality assurance: Audit box maps periodically and resolve mismatches immediately. Protect high-value stocks. Use locked boxes or controlled access for irreplaceable or sensitive materials, and track who has access. 
- (6.) Agree on what “validated” means for each material type before adding it to the inventory. For example, mycoplasma-negative cell lines, sequence-confirmed plasmids, spectroscopically pure compounds.

>> **General procedure: Banking a new material**

- (1.) Pull the next identifier from the inventory.
- (2.) Label the tube or container in water-resistant ink or on a cryo-safe label. Frozen plastic doesn't hold ink or stickers well.
- (3.) Record all required metadata in the shared inventory at the time of banking. At a minimum: plate/well ID, vector, host, passage, selection conditions, date, phenotype notes, and analysis method. 
- (4.) Prepare multiple vials or aliquots whenever feasible, while the material is healthy and not after it has aged or been stressed. Store at least one in a backup location such as a separate freezer, with a collaborator, or in a repository.
- (5.) Place the vials into indexed cryoboxes or storage racks and update the digital box map immediately.

Banking of continuously renewable materials

- (1.) Maintain at least one backup aliquot for each material under long-term storage conditions. Keep prior batches for at least two years to enable troubleshooting and reproducibility checks.
- (2.) Withdraw a sample from the remaining working stock for your experiments.
- (3.) When renewing a stock, assign a new identifier and validate its quality before adding it to the inventory.

Banking of cell lines

- (1.) Prepare stocks for the master cell bank from early-passage, validated cultures. The master bank is never thawed except to produce working stocks. 
- (2.) From the master bank, expand working cell banks for routine use. Store each working bank separately from the master so that a single freezer failure does not lose both. 
- (3.) *Critical:* Validate each working batch for identity (STR profile), absence of contamination (mycoplasma testing), and phenotype before banking.
- (4.) Thaw and expand one vial from the working cell bank for your experiments.

Banking of limited, single-batch materials

- (1.) Record the total initial quantity at banking and a “critical low” threshold in the inventory. 
- (2.) Aliquot into single-use vials where possible to avoid freeze-thaw cycles and contamination.
- (3.) Update the inventory after each use to reflect the exact remaining amount. Flag the record when the critical low threshold is reached.
- (4.) At depletion, mark the record as exhausted and archive it for traceability.

>> General procedure: Retrieving and using a banked material

- (1.) Locate the vial in the inventory database and verify its identifier before retrieval.
- (2.) Log the user, date, and purpose of retrieval in the inventory system.
- (3.) After use, return the vial if still viable, or update the inventory to mark it as depleted or disposed.

Critical: Never return thawed cell lines or degraded reagents to long-term storage.

>> Disposal and retirement of stocks

- (1.) When a material is depleted, contaminated, fails validation, or is no longer needed, mark it as “retired” in the inventory database; log the date and the reason for disposal. Do not erase the record.
- (2.) Physically remove the vial(s) from storage and dispose of them following the appropriate biosafety or chemical-waste disposal protocol.

>> Material sharing and chain-of-custody

- (1.) When transferring any material to another person, lab, or collaborator, record the stock identifier, recipient, date, and purpose in the inventory database.
- (2.) Where applicable, ensure that a Material Transfer Agreement (MTA) or other legal document is in place before the transfer.

Critical: Do not share materials subject to export control, clinical privacy, or other restrictions without prior approval.

- (3.) Shipment of hazardous or biological materials may need to be reviewed by EHS. For BSL-2 materials, confirm that the recipient has appropriate facilities and training. 

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